The Function of Villi in the Small Intestine

Target Subject: Life science/biology

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Author

Sarah Hughes, Head of Science, New College Worcester, United Kingdom

Purpose

To examine a model of the inside of the small intestine to support understanding of the function of villi in digestion

Background information



The structure of the small intestine is designed for absorption of nutrients. The inside of the small intestine is lined with villi that absorb nutrients from the liquid mixture called chyme produced in the stomach from the food we eat. Biology books usually have a detailed picture of the villi. Some plastic science models also show villi. However, this easy to create model more accurately presents the concept of villi in a way that can be accessed tactually and visually. Blood vessels on the outside of the intestinal wall absorb nutrients so that they can travel around the body in the blood.

Preparation

Use a bath mat made of chenille or other material with loops, cover the outside of the rug (non-looped side) with mesh (preferably red) left over from a bag of onions from the grocery store or other source. The mesh will replicate the blood vessels. Roll the rug into a tube with the loops on the inside. Fasten the rug with Velcro or hook and eye closures.

Materials

Small bath mat, approximately 2ft by 3ft.
Mesh or netting from an empty bag of onions, or other mesh or netlike material
Velcro or hook and eye closures
Tactile drawing of villi and small intestine

Procedure

1. After reading about the digestive system and the structure and function of the small intestine, students examine tactile pictures and models of the small intestine and the villi.
2. Place hand or arm and hand inside the tube shaped intestine model. Notice the villi touching your arm and hand.
3. Examine the mesh netting simulating the small blood vessels on the outside of the intestine. Remember from your reading that the blood vessels in the wall of the intestine absorb the nutrients from food as it passes through the small intestine.
4. Open the "intestine" to more closely examine the villi. Notice how the villi create an increased surface area to maximize the absorption of nutrients.

Resources

Hughes, Sarah, Head of Science. Making Models: New College Worcester, United Kingdom, 2009.
Miller, Kenneth R. PhD. And Joseph s. Levine, Biology: Foundation Edition: Pearson Education, Inc., 2010.

About Sarah Hughes

As the Head of Science at New College Worcester (NCW), Sarah Hughes is a Qualified Teacher of Visually Impaired learners.

NCW is a Specialist School for Visually Impaired young people (aged 11 to 19 years) in Worcester, England. Sarah joined the school from a large mainstream High School in September 2007. When offering advice to teachers and support staff of visually impaired students she is always conscious of the fact that things are very different for a teacher in a specialist setting compared to one in a mainstream classroom with 30 students (and only one visually impaired student).

In mainstream schools many teachers are regularly using technology to show animations and interactive whiteboard presentations. Since learning to teach visually impaired students, Sarah has introduced and developed the use of models in teaching and learning. These successfully complement the use of ICT in her classroom.